

CSE 350

Digital Electronics and Pulse Techniques

Digital to Analog Converter (DAC)

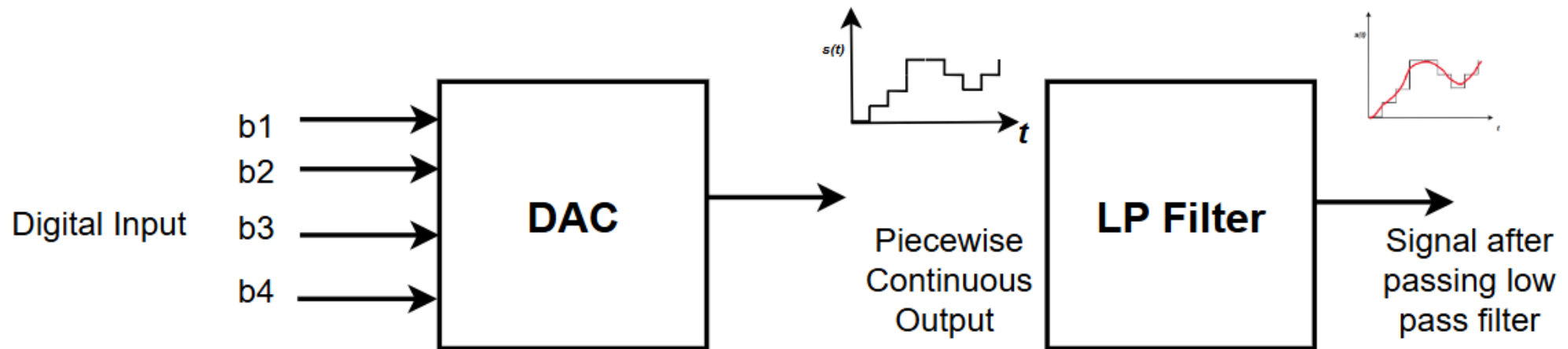
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Inspiring Excellence

Digital to Analog Converter (DAC)



Binary to Decimal Conversion

Binary Number: $B_3B_2B_1B_0$

Equivalent Decimal Number: $B_3 * 8 + B_2 * 4 + B_1 * 2 + B_0 * 1$

Example : Convert 1101_2 value into a decimal value.

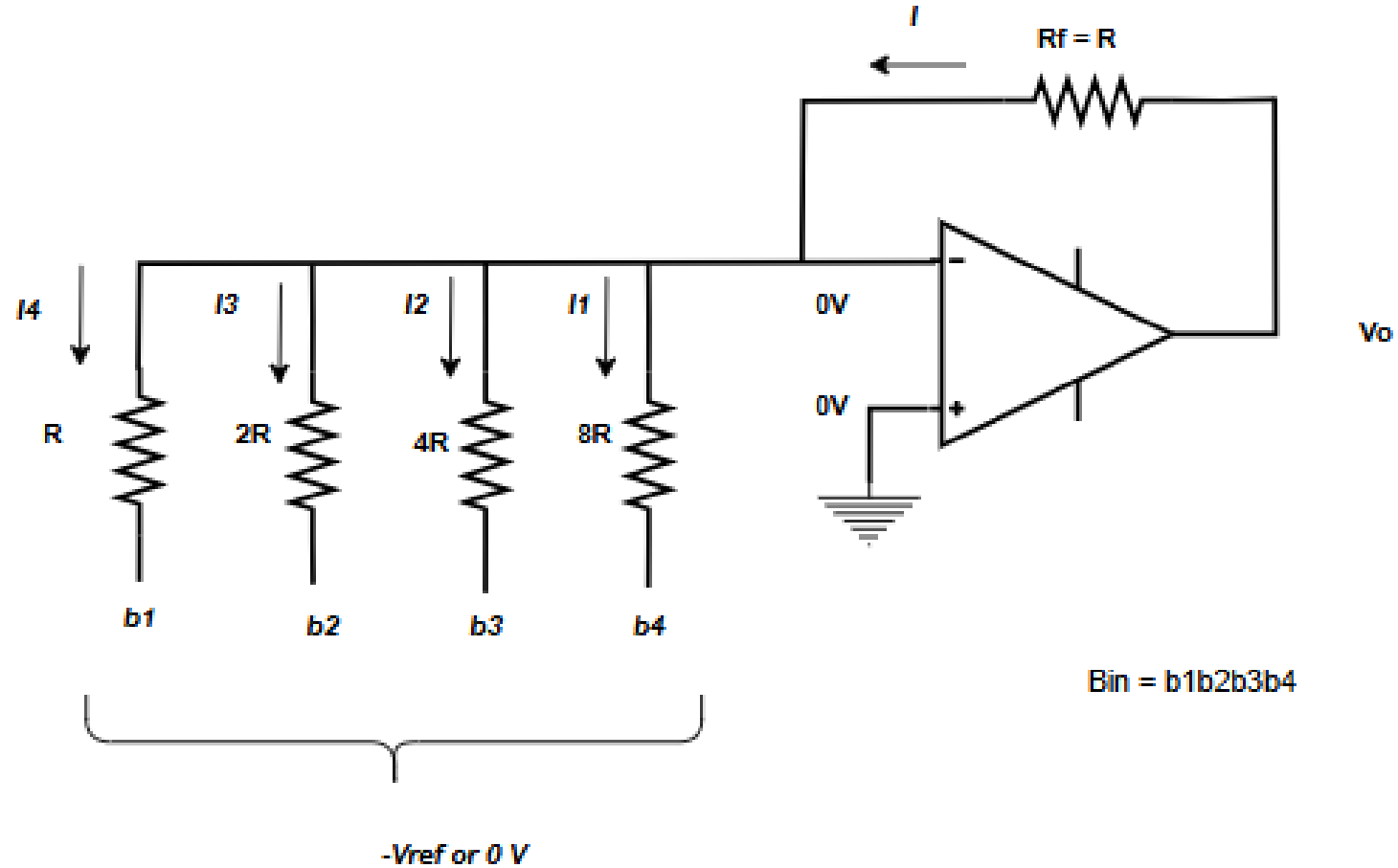
Solve: $1001_2 = 1 * 8 + 0 * 4 + 0 * 2 + 1 * 1 = 9_{10}$

Binary Weighted DAC

Key features of this circuit,

- LSB branch will contain biggest resistor
- MSB branch will contain Smallest resistors
- Output will be positive if the input voltage is negative
- Resistor of adjacent branch will be either double or half
- There are 2^n output level for n bit input.

Output of the DAC circuit



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If $b_1 = 1$, input of first branch is connected to the $-V_{REF}$ and

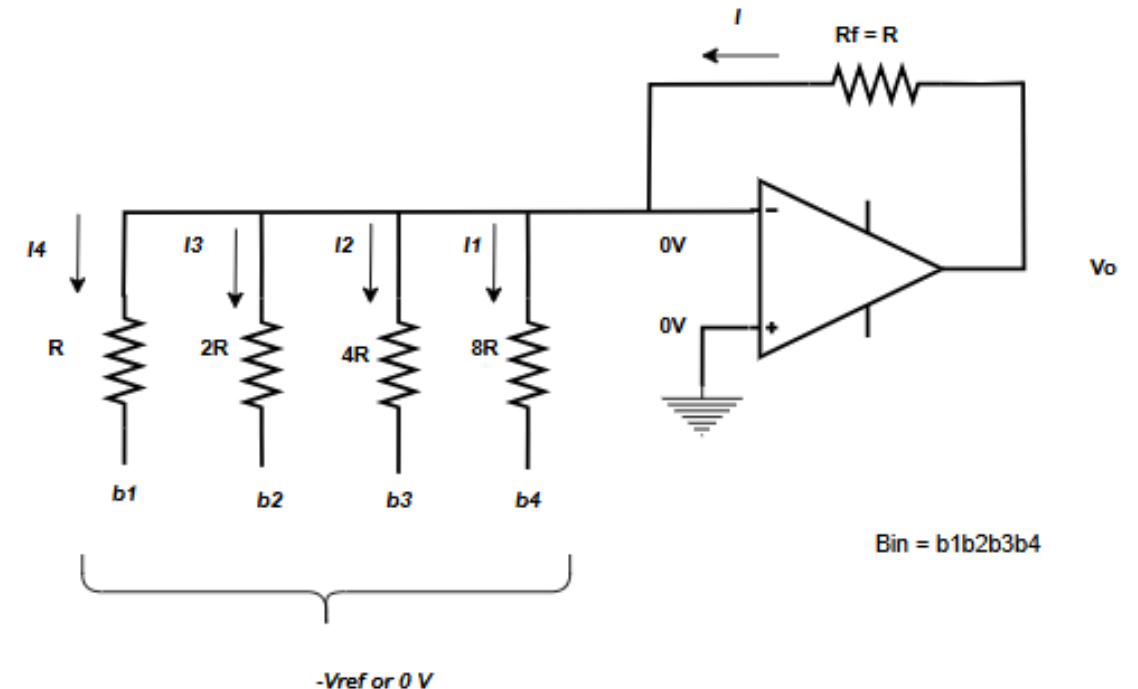
If $b_1 = 0$, input of the first branch is connected to the $0V$ (GND).

Same goes for the b_2, b_3, b_4

Voltage of the first branch can be modeled as,

$$V_1 = -b_1 V_{REF}$$

Same for other brach.



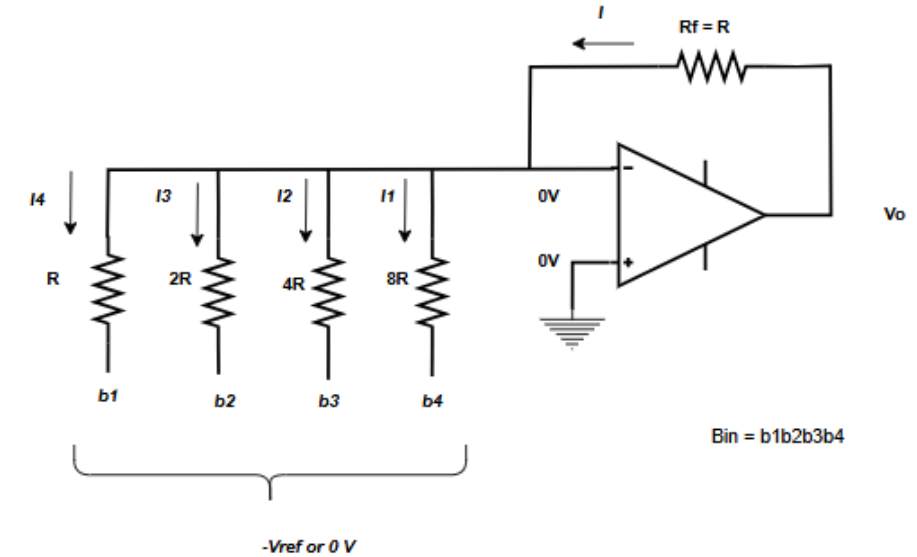
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Node equation at V_- terminal,

$$I = I_1 + I_2 + I_3 + I_4$$

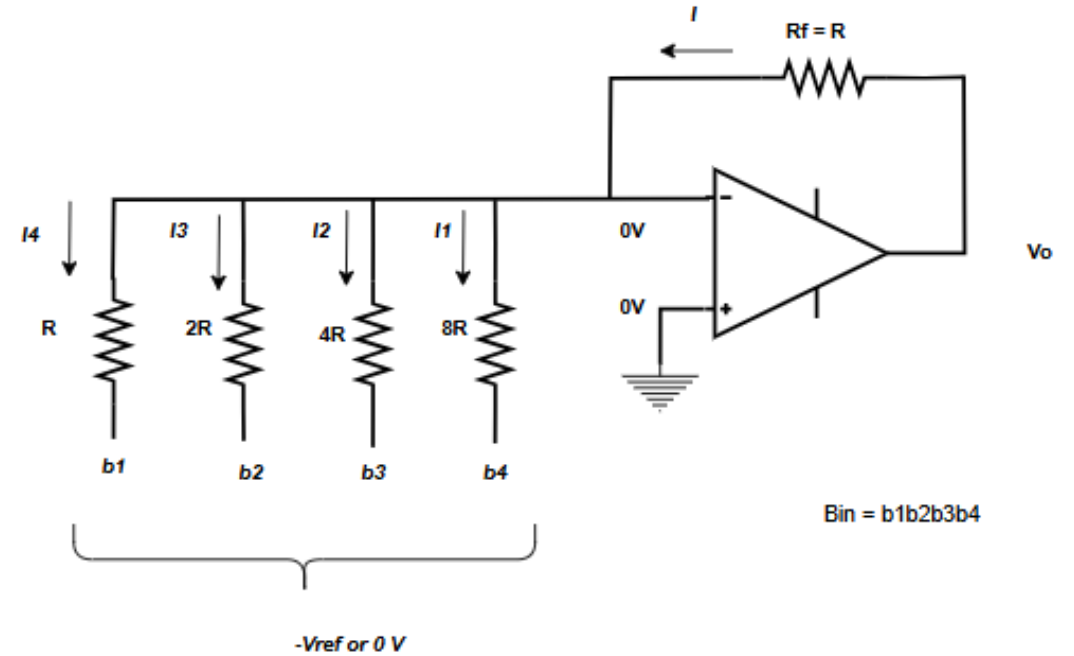
$$\Rightarrow \frac{V_o - 0}{R} = \frac{0 - (-b_1 V_{REF})}{R} + \frac{0 - (-b_2 V_{REF})}{2R} + \frac{0 - (-b_3 V_{REF})}{4R} + \frac{0 - (-b_4 V_{REF})}{8R}$$

$$V_o = V_{REF} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right)$$



Digital to Analog Converter (DAC)

Table: For different input combination.
Combination = $2^4 = 16$, $V_{REF} = 8\text{ V}$



b1	B2	b3	b4	Vo
0	0	0	0	0V
0	0	0	1	1V
0	0	1	0	2V
0	0	1	1	3V
0	1	0	0	4V
0	1	0	1	5V
0	1	1	0	6V
0	1	1	1	7V
1	0	0	0	8V
1	0	0	1	9V
1	0	1	0	10V
1	0	1	1	11V
1	1	0	0	12V
1	1	0	1	13V
1	1	1	0	14V
1	1	1	1	15V



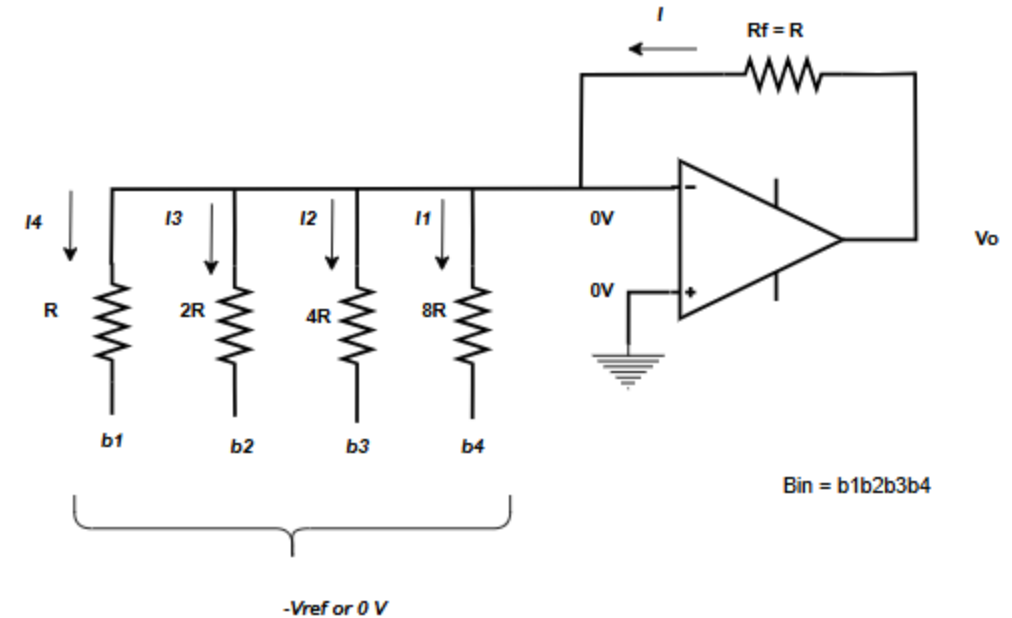
Step Size

If $V_{REF} = 8\text{ V}$, what will be the step size ?

Step size is the difference between two adjacent output values or the output value for the LSB bits contribution only.

For 0001 input, output will be the step size.

$$\Delta = V_{REF} \left(b1 + \frac{b2}{2} + \frac{b3}{4} + \frac{b4}{8} \right)$$
$$= 8 \left(0 + \frac{0}{2} + \frac{0}{4} + \frac{1}{8} \right) = 1\text{ V}$$



Maximum Output

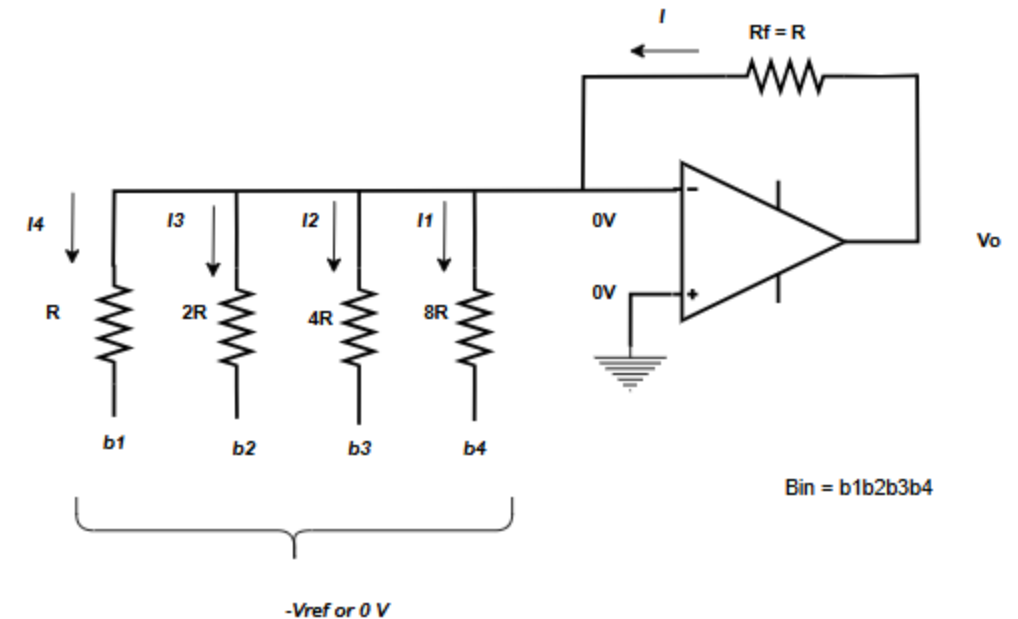
If $V_{REF} = 8\text{ V}$, what will be the

Max output ?

Max output level can be found for high input at every input.

For 1111 input, output will be max.

$$\Delta = V_{REF} \left(b1 + \frac{b2}{2} + \frac{b3}{4} + \frac{b4}{8} \right)$$
$$= 8 \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) = 15\text{ V}$$



General Formula for DAC

Binary: DCBA

$$V_o = -R_f \left(\frac{V_D}{R_D} + \frac{V_C}{R_C} + \frac{V_B}{R_B} + \frac{V_A}{R_A} \right)$$

$R_A > R_B > R_C > R_D$ and factor of 2

If we assign,

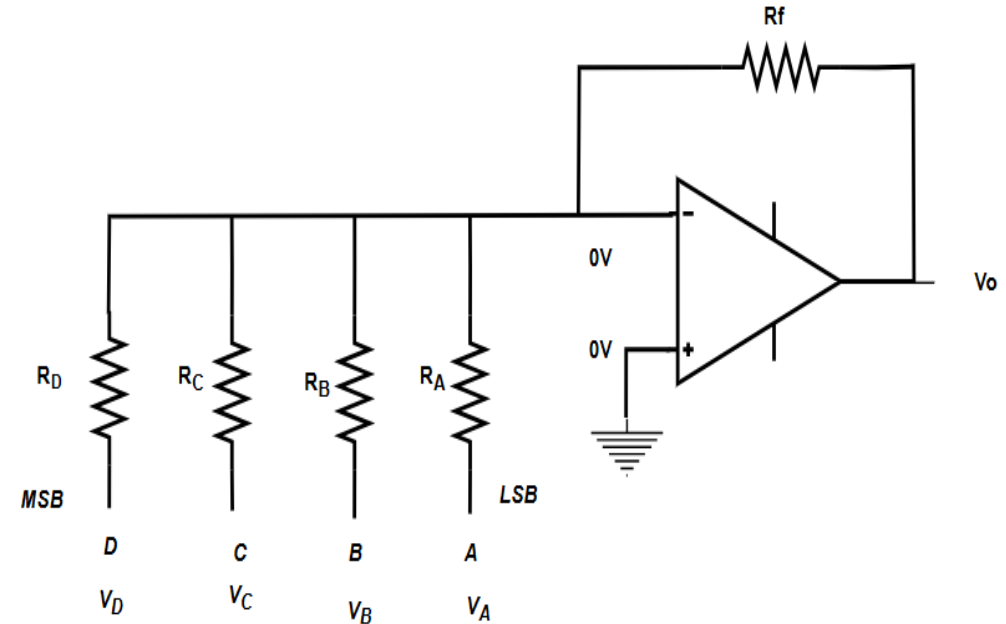
$R_D = R$, then, $R_C = 2R$, $R_B = 4R$, $R_A = 8R$

If we assign,

$R_A = R$, then, $R_B = R/2$, $R_C = R/4$, $R_D = R/8$

If we want, positive output, V_{REF} need to be negative.

If we want, negative output, V_{REF} need to be positive.



DAC (MSB $\rightarrow$ R)

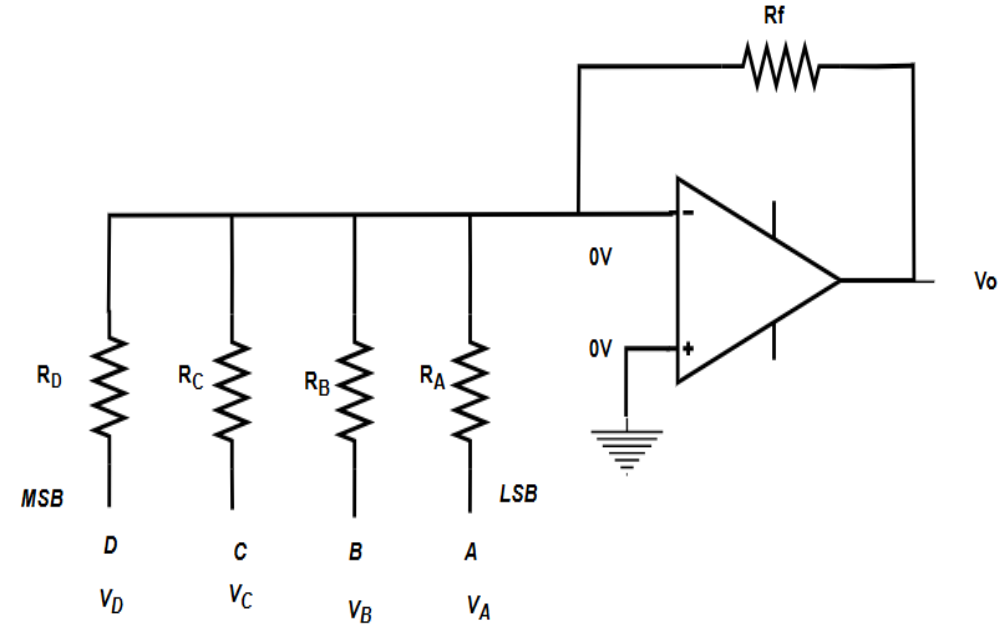
Binary: DCBA

If $V_A = A * V_{REF}$, $V_B = B * V_{REF}$,
 $V_C = C * V_{REF}$, $V_D = D * V_{REF}$

$R_D = R$, then, $R_C = 2R$, $R_B = 4R$, $R_A = 8R$

$$V_o = -\frac{R_f}{R} \left(V_D + \frac{V_C}{2} + \frac{V_B}{4} + \frac{V_A}{8} \right)$$

$$V_o = -\frac{R_f * V_{REF}}{R} \left(D + \frac{C}{2} + \frac{B}{4} + \frac{A}{8} \right)$$



DAC (LSB $\rightarrow$ R)

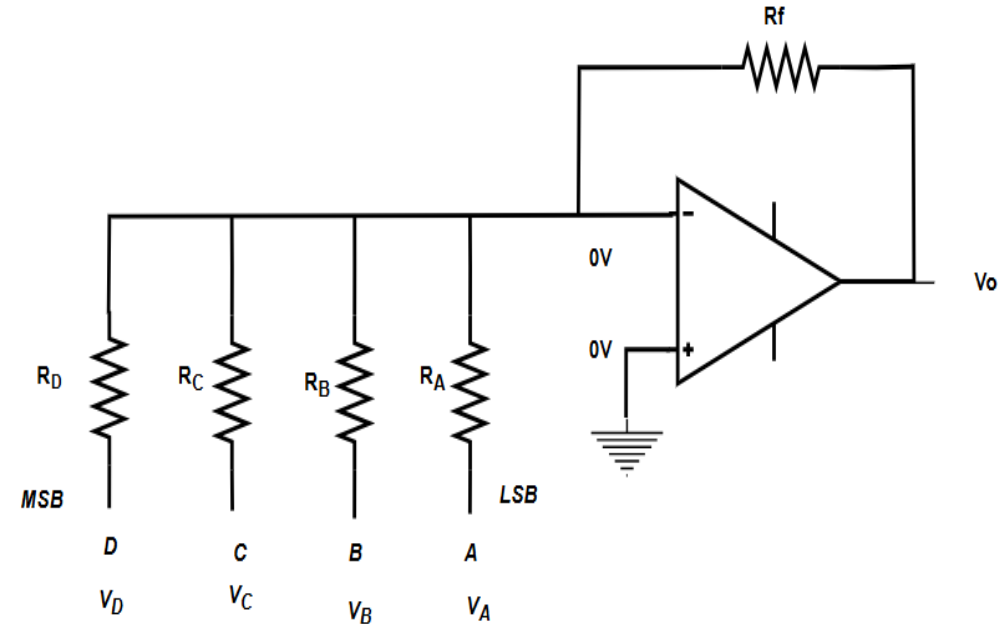
Binary: DCBA

$$\text{If } V_A = A * V_{REF}, V_B = B * V_{REF}, \\ V_C = C * V_{REF}, V_D = D * V_{REF}$$

$$R_A = R, R_B = \frac{R}{2}, R_C = \frac{R}{4}, R_D = \frac{R}{8}$$

$$V_o = -\frac{R_f}{R} (8 * V_D + 4 * V_C + 2 * V_B + 1 * V_A)$$

$$V_o = -\frac{R_f * V_{REF}}{R} (8 * D + 4 * C + 2 * B + 1 * A)$$

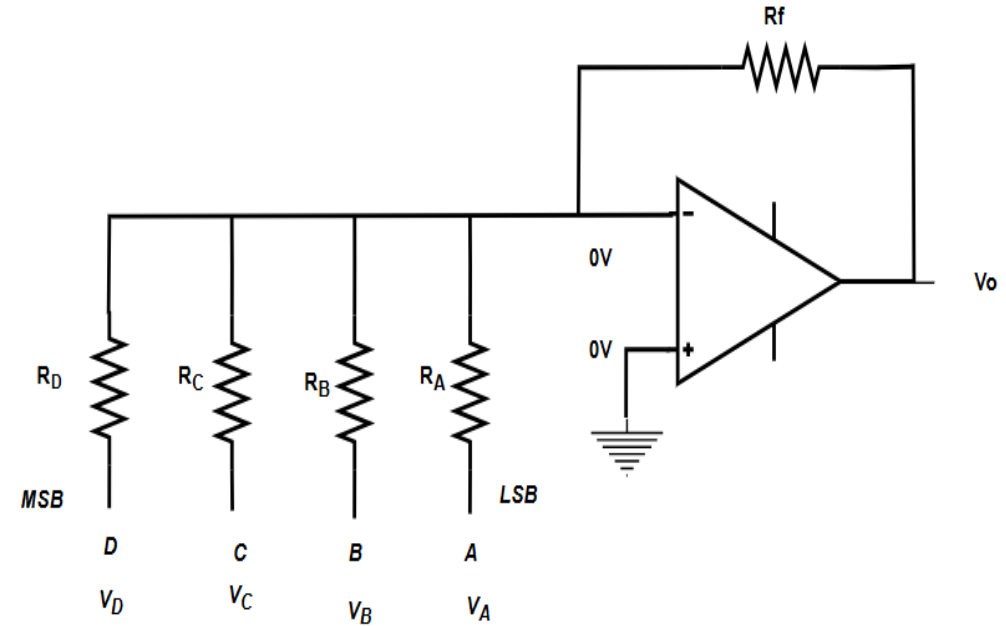


Exercise:

Binary: DCBA

If $R_A = R_F = 10k\Omega$, Find the following parameter for this DAC circuit. Given, $V_{REF} = 0.5\text{ V}$, $V_H = 30\text{ V}$ and $V_L = -30\text{ V}$

- Find the value of maximum output.
- Find the value of the step size.
- Find the value of the 1LSB voltage.
- Find the value of the 1MSB value.
- Plot the Output voltage vs input binary.



Exercise:

The figure below shows the input-output characteristic of a converter. The output of the converter is denoted as V_O and the input as V_{in} . Identify the converter type and the number of bits used by the converter. Further calculate V_{ref} when $R_F = 2R$.

